

StickForStats

Developer Handover & Scientific Design Contract

(Authoritative document for future development)

1. Executive Summary (What This Software Is and Why It Matters)

StickForStats is **not** a statistical library, a GUI wrapper, or a convenience tool. Its core contribution is **methodological and architectural**:

StickForStats enforces statistical assumption awareness as a first-class object in every analysis workflow.

This means:

- Statistical results are **never detached** from their assumptions.
- Assumption validation is **automatic, explicit, and unavoidable**.
- The system is designed to reduce silent misuse of statistical tests, a known contributor to irreproducible research.

Any future development **must preserve this philosophy**, or the software loses its scientific value and publishability.

1.1 What StickForStats Is

- A **statistical analysis platform** with mandatory assumption validation
 - A **methodological intervention** in statistical practice
 - A **research-grade system** intended for publication and reuse
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1.2 What StickForStats Is Not

- **✗** A black-box statistics engine
- **✗** A “click-and-get-p-values” tool
- **✗** A UI-first analytics dashboard
- **✗** A replacement for statistical reasoning

Automation in StickForStats exists **to expose assumptions**, not to hide decisions.

1.3 Central Design Principle (Non-Negotiable)

No statistical result may exist without an explicit, traceable assumption context.

This principle overrides:

- UI convenience
- performance optimizations
- feature additions
- AI integrations

If a change violates this principle, it must **not** be merged.

2. Core Scientific Design Principles (Non-Negotiable)

This section defines **hard constraints**.

All contributors must treat these as a **design contract**, not suggestions.

2.1 Guardian Is the Scientific Core

The **Guardian system** is the epistemic backbone of StickForStats.

Guardian is responsible for:

- Identifying which assumptions apply to a statistical test
- Validating those assumptions automatically
- Reporting violations with severity
- Producing a confidence score
- Recommending alternative analyses when appropriate

Invariant (Must Always Hold)

Every statistical test execution must invoke Guardian.

There are **no exceptions**.

2.1.1 Required Guardian Outputs

For every test execution, Guardian must return:

- `assumptions_checked`
- `violations` (if any)

- `severity_levels`
- `confidence_score`
- `alternative_test_recommendations`

If any of these are missing, the execution path is **scientifically invalid**.

2.1.2 Adding New Statistical Tests

When adding a new test (e.g., survival analysis, mixed models):

The developer **must**:

1. Declare required assumptions in the Guardian registry
2. Implement validators **or**
3. Explicitly document assumptions that cannot be tested automatically

Silence is not allowed.

“Not checked” must be explicit.

2.2 Non-Blocking by Design (But Not Optional)

Guardian **must not silently block** analyses by default.

Rationale

- Statistical assumptions are **context-dependent**
- Violations exist on **continua**, not binaries
- Domain expertise may justify violations

Blocking is a **policy decision**, not a statistical one.

2.2.1 Correct Behavior

Guardian **must**:

- Warn
- Score
- Document
- Recommend alternatives

Guardian **must not**:

- Silently proceed
 - Silently block
 - Decide “truth” for the user
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2.2.2 Acceptable Exception: Configurable Strict Mode

A strict / protected mode **may exist**, but only if:

- It is **explicitly enabled**
- It is **clearly documented**
- It does **not** alter the default scientific narrative

The default behavior remains **informative, not authoritarian**.

2.3 Determinism Is Mandatory for Scientific Results

Anything that influences **numerical results, validation, or reporting** must be:

- Deterministic
- Reproducible
- Independent of external services

This includes:

- Assumption checks
 - Test execution
 - Confidence scoring
 - Validation scripts
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2.3.1 AI Is Optional and Non-Authoritative

AI systems may:

- Explain concepts
- Assist users
- Draft text

AI systems must **never**:

- Validate assumptions
- Compute statistics
- Influence Guardian outputs
- Affect reproducibility

If AI is unavailable, **scientific results must remain unchanged**.

2.4 Extensibility Without Scientific Drift

The system must be extensible **without weakening guarantees**.

Acceptable extensions:

- New validators
- New test types
- New visual diagnostics

Unacceptable extensions:

- Bypassing Guardian
 - Hiding assumption failures
 - Replacing statistical logic with heuristics or AI
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2.5 Developer Decision Rule (Use This Before Every Change)

Before merging any change, ask:

1. Does this preserve assumption transparency?
2. Does it preserve or improve reproducibility?
3. Can a statistician reviewer understand and defend it?
4. Does it avoid silent automation?

If **any answer is “no”**, the change must be reconsidered.

Closing Note to Developers

StickForStats is designed to **raise the floor of statistical practice**, not to lower the bar of responsibility.

Automation here is **protective**, not permissive.

Treat Guardian as a **scientific object**, not just code.
